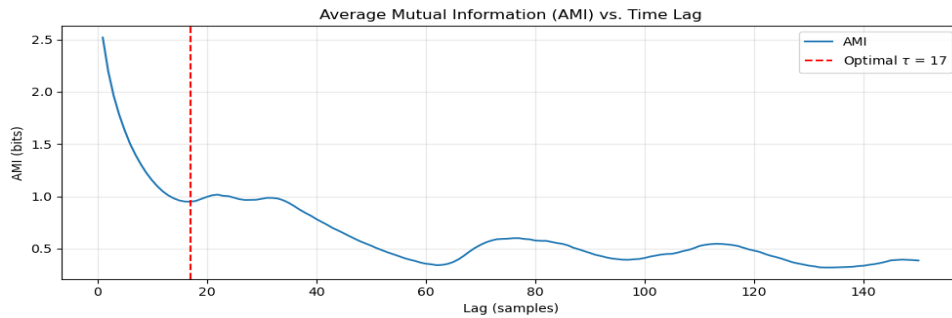


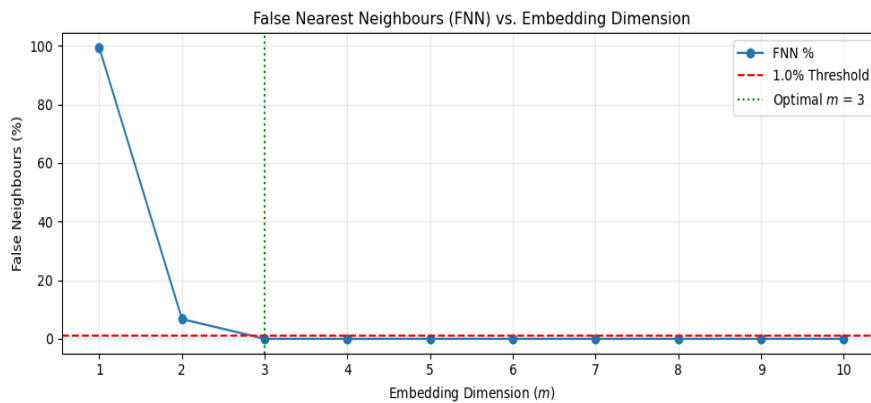
Task (2) Report “Evaluating Systems”

Part A: (Phase Space Reconstruction)

- A.1) AMI plot (optimal $\tau = 17$)



- A.2) FNN% plot (Optimal $m = 3$)



- Comparing the Lorenz Dimensions to the calculated Embedding Dimensions:

Known Lorenz State Dimensions	Calculated Embedding Dimensions (m)
3	3
Lorenz system is defined by three ODEs (x,y,z) so it has 3 state dims.	The estimated “m” is enough to expand and unfold the system data.

Part B: (Lyapunov Exponent)

B.1) Spectrum from “Lyapunov Wolf ODE”:

- Lyapunov ODE Spectrum (bits/time): [1.29482562e+00, -3.80432195e-03, -2.10077070e+01]
- Largest Lyapunov Exponent (λ_1) “ground truth”: 1.2948 bits/time

B.2) λ_1 From “Wolf MLE” on Scalar Time Series:

- Wolf-MLE Lyapunov Exponent: 2.2995 bits/time
- Tracked-Segments (replacements): 299

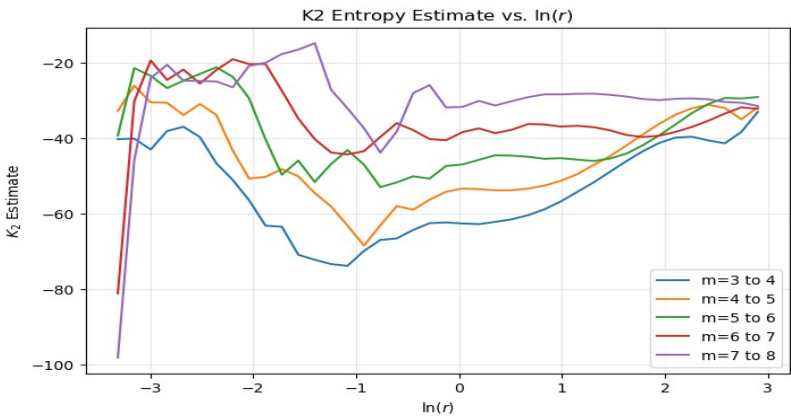
λ_1 Lyapunov Wolf ODE	λ_1 Wolf MLE
1.2948 bits/time	2.2995 bits/time

- The reconstructed data is more chaotic (harder to predict) having larger value of Lyapunov exponent (as Wolf MLE applied on the reconstructed data), as it is less clean than the directly generated data from Lorenz equation.

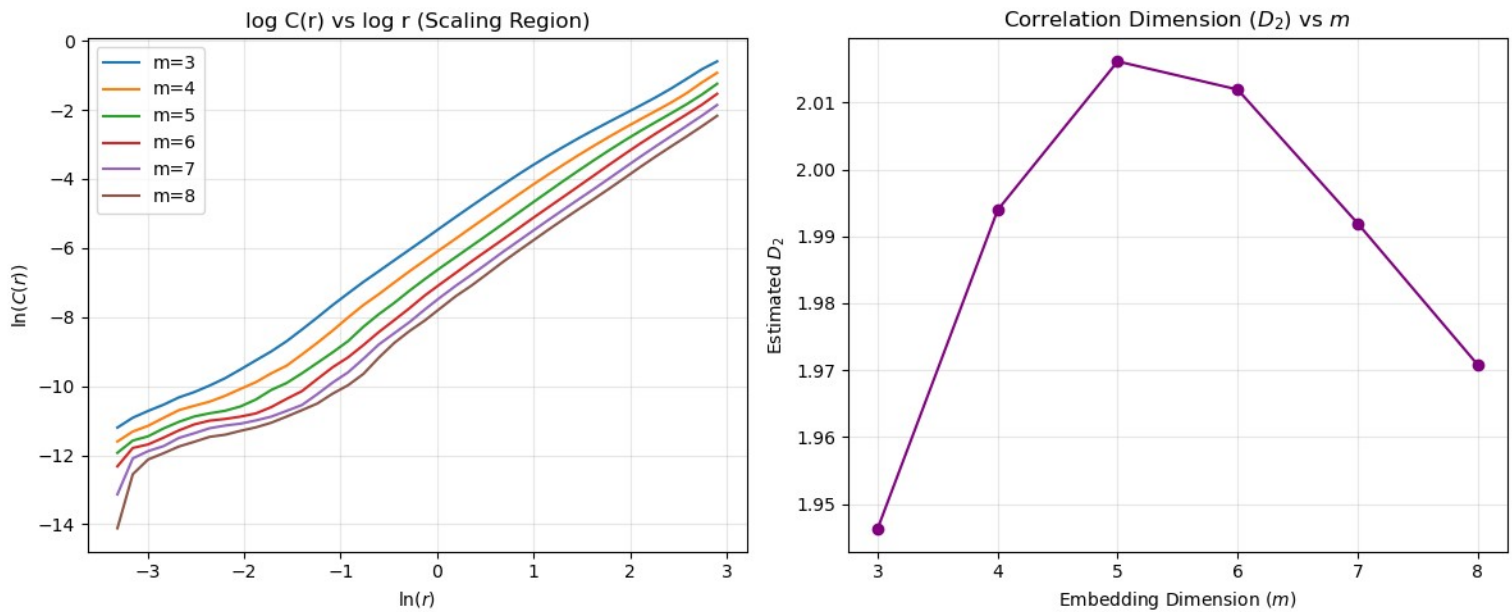
Part C: (K2 Entropy Estimate)

C.1) K2 Entropy Estimate:

- K2 entropy approximately equals -37 for (m=6 to 7)
- Yes, The K2 appears to saturate as the embedding dimension increases, the curves converge vertically and form a stable horizontal plateau in the scaling region, indicating deterministic low-dimensional chaos rather than random noise.
- But it saturates to a negative value. (but I think it is a typo in the K2 formula in the task guide “having additional negative sign”)

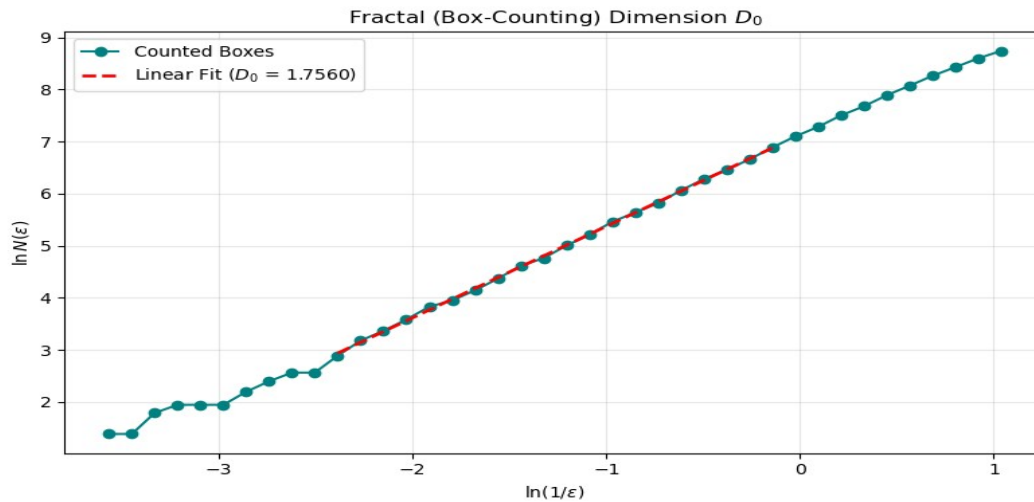


Part D: (Correlated Sum “C_m(s)” & Correlation Dimension “D2”



- D_2 curve is not plateauing perfectly because running out of data. This is a well-known phenomenon in phase space reconstruction called **data depletion**.
- But the D_2 value saturation point is at the peak of the curve (about 2.015) corresponding to ($m = 5$)
- The subsequent drop at $m > 5$ is a known property of finite series length (data depletion), where the extreme sparsity of the high-dimensional space distorts the linear scaling region.

Part E: [Fractal (Box-Counting) Dimensions]



- Extracted $D_0 = 1.7560$, but the $D_2 = 2.015$
- While theory states $D_0 \geq D_2$, the box-counting method (D_0) severely underestimates the dimension in practice because our limited dataset leaves most of the tiny microscopic boxes entirely empty. The correlation method (D_2) is statistically much more accurate and robust because it evaluates millions of actual pairwise distances instead of relying on an artificial grid.

Final Conclusion:

- The simulated Lorenz system is classified as chaotic system, neither periodic nor hyperchaotic, because its full Lyapunov spectrum resulted in only one positive exponent. This classification of low-dimensional, deterministic chaos is strongly proofed by these metrics: the K2 entropy saturates to a finite positive value (the negative value I got is highly probably a typo in the given formula), and the correlation dimension (D_2) converges to a non-integer fractal dimension rather than growing boundlessly with the embedding space."